

Build4All Product Events and Conversion Policy

Controlled knowledge source for multilingual Text to SQL retrieval

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Version	2.0
Authoritative schema	build4all_research
Status	Controlled research source
Scope	Recorded product events and view-to-purchase conversion

Purpose and authority

This document defines product-event counting and view-to-purchase conversion for the frozen build4all_research PostgreSQL schema. It is authoritative for the controlled retrieval experiment and must remain synchronized with the benchmark gold queries.

Authoritative event model

Each row in product_events represents one recorded event. The event category is stored in product_events.event_type. Count event rows directly; do not infer product events from orders, order_items, product stock, product status, or customer counts.

Event count by type

To report product-event volume, group product_events rows by event_type and count every row in each group. Preserve every event type present in the table. Do not restrict the result to views or purchases, and do not add product, customer, order-status, or date filters unless the request explicitly asks for them.

An event count is a row count. It is not a distinct-customer count, distinct-product count, order count, or quantity sold. Use customer or product uniqueness only when the question explicitly requests it.

View-to-purchase conversion

The controlled conversion rate compares recorded purchase events with recorded view events across the requested scope. Its formula is 100 multiplied by the number of rows whose event_type is purchase, divided by the number of rows whose event_type is view.

$$\text{conversion percentage} = 100 \times \text{purchase-event count} \div \text{view-event count}$$

Use a null-safe denominator so division by zero cannot occur. When the view-event count is zero, the rate is undefined and should remain null unless an approved reporting requirement explicitly defines another display value. Do not divide purchases by all events, customers, products, orders, or order-item quantity.

SQL semantic pattern

Conditional aggregation should count purchase and view rows from the same product_events population. A PostgreSQL implementation may use COUNT(*) FILTER conditions and NULLIF around the view count. Keep the numeric multiplier decimal so the percentage is not truncated by integer division.

Scope and grouping rules

- For a global conversion question, aggregate the full requested event population without grouping by product, customer, or date.
- Group by product, customer, date, or event type only when that dimension is explicitly requested.
- Apply a time condition to the product-event timestamp only when the question specifies an event period.
- Do not borrow order dates or order statuses for a product-event metric unless the request explicitly combines those concepts.
- Do not join other tables when product_events alone contains every field needed for the requested metric.

Validation safeguards

- Verify that the event_type comparison uses the exact stored values for view and purchase events.
- Verify that numerator and denominator use the same filters and scope apart from event type.
- Verify that event counts are row counts and are not silently deduplicated.
- Verify that the zero-view case is protected from division by zero.
- Verify that no unrelated status, product, customer, date, or inventory filter was added.

Missing data and interpretation

The metric describes recorded product events only. An absent row does not prove that an interaction never occurred outside the tracked system. Do not invent sessions, unique visitors, channels, devices, attribution sources, or deduplication rules when those definitions are unavailable.

Research governance

This is a synthetic controlled research policy, not an audited production analytics policy. Changes to event values, counting rules, conversion scope, or zero-denominator handling require a new version and synchronized benchmark updates. Benchmark questions and expected result values are intentionally excluded from this retrieval document.